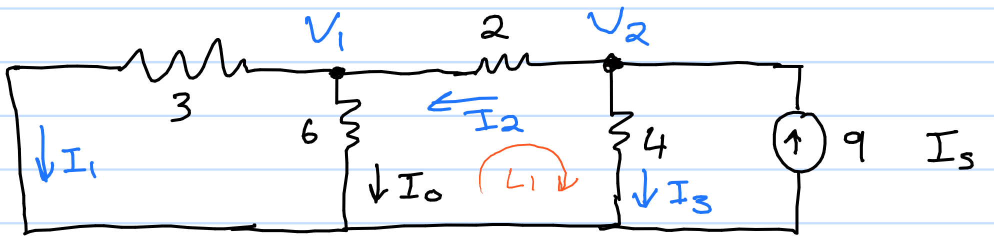
4.4 Linearity property

Using linearity determine I_o



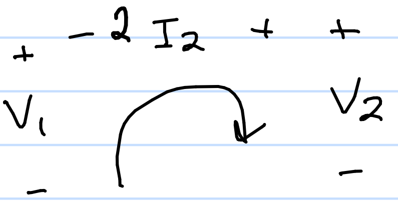
Assume $I_o = 1$ A (Linearity)

$V_1 = 6 I_o = 6$ V (Ohm's law)

$I_1 = \frac{V_1}{3} = \frac{6}{3} = 2$ A (Ohm's law)

$I_2 = I_1 + I_o$ (KCL at V_1)
$= 2 + 1$
$= 3$ A

(KVL at L_1)



$-V_1 - 2 I_2 + V_2 = 0$

$V_2 = V_1 + 2 I_2$
$= 6 + 2 \times 3$
$= 12$ V

$I_3 = \frac{V_2}{4}$ (Ohm's law)

$= \frac{12}{4}$

$= 3$ A

$I_s = I_2 + I_3$ (KCL at V_2)
$= 3 + 3$
$= 6$ A

This gives $I_s = 6 I_o$ (Linearity)

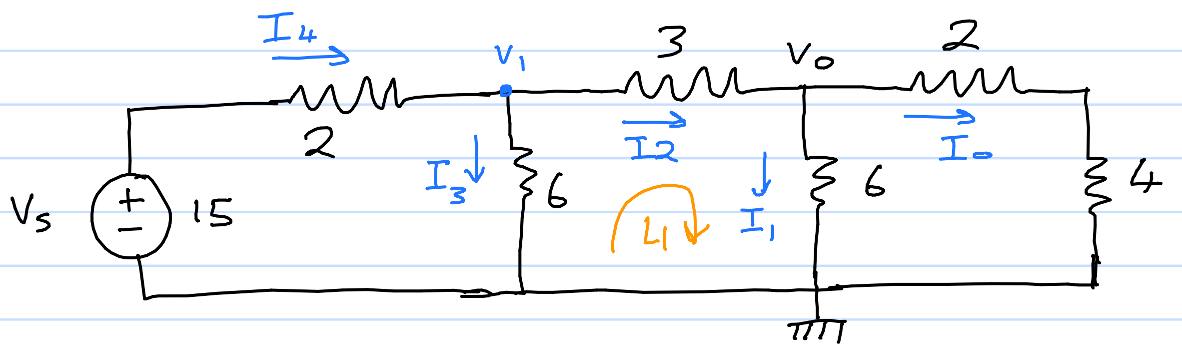
$I_o = \frac{I_s}{6}$

$= \frac{9}{6}$

$= 1.5$ A

4.5 Linearity property

Assume $V_o = 1$ V, use linearity to find V_o



Assume $V_o = 1$ V

$I_o = 1/6$ $\quad I_1 = 1/6$ (Ohm's law)

$I_2 = I_1 + I_o = 2/6$ (KCL at V_o)

$V_1 = V_o + 3I_2$ (KVL at L_1)

$= 1 + 3 \times 2/6$

$= 2$ V

$I_3 = 2/6$

$I_4 = I_3 + I_2$ (KCL at V_1)

$= 2/6 + 2/6$

$= 4/6$

$$I_4 = \frac{V_s - V_1}{2}$$ (Ohm's law)

$$\frac{4}{6} = \frac{V_s - 2}{2}$$

$4 = 3V_s - 6$

$$V_s = \frac{10}{3}$$

$$V_s = \frac{10}{3} V_o$$ (Linearity)

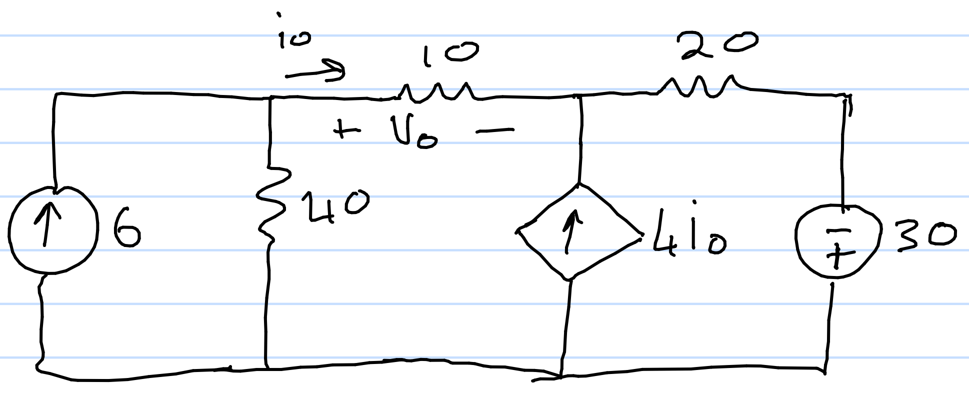
$$V_o = \frac{V_s\,3}{10}$$

$$= \frac{15 \times 3}{10}$$

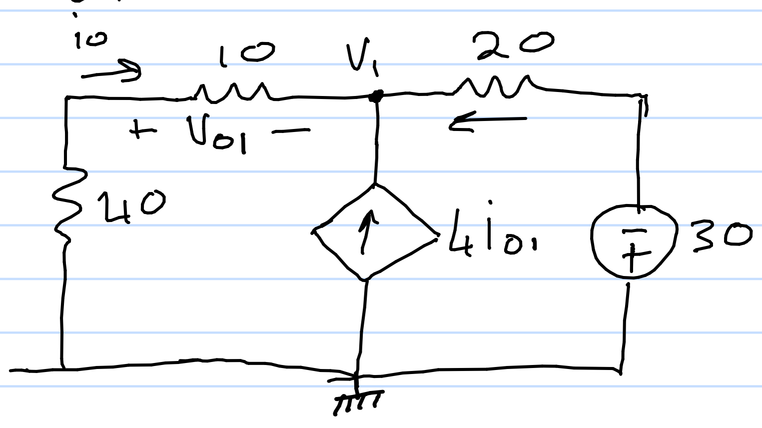
$= 4.5$ V

4.11 Super position

Use superposition to find V_o & i_o



Switch of 6 A source



KCl at V_1

$$i_{o1} + 4i_{o1} + \frac{(-30) - V_1}{20} = 0 \qquad (1)$$

Ohm's law at V_1

$$V_1 = -(40+10)\, i_{o1} = -50 i_{o1} \qquad (2)$$

(Voltage always drops when current flows through resistor, V_1 is a drop from ground)

Substitute 2 into 1

$$5i_o + \frac{(-30) + 50i_{o1}}{20} = 0$$

$$100 i_o - 30 + 50 i_{o1} = 0$$

$$150 i_{o1} = 30$$

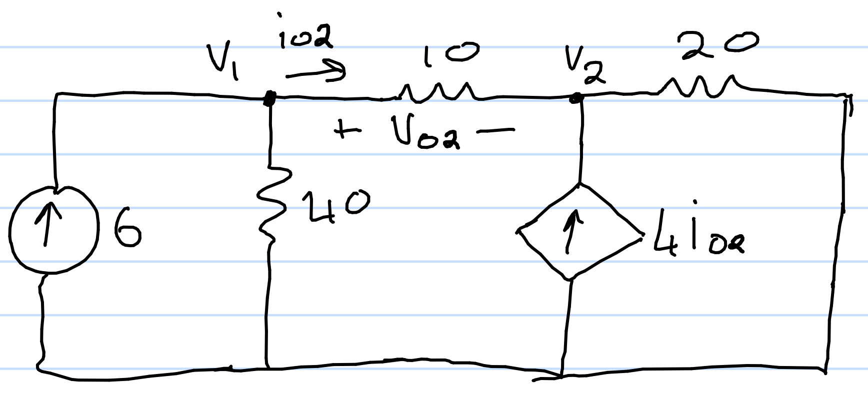
$$i_{o1} = 0.2 \text{ A}$$

Ohm's law at V_o

$$V_{o1} = 10 i_{o1}$$

$$= 2 \text{ V}$$

Switch of 30 V source



$$V_o = V_1 - V_2$$

Perform nodal analysis

$$-6 + \frac{V_1}{40} + \frac{V_1 - V_2}{10} = 0$$

$$V_1 + 4V_1 - 4V_2 = 240$$

$$5V_1 - 4V_2 = 240 \qquad (1)$$

$$\frac{V_2 - V_1}{10} - 4i_o + \frac{V_2}{20} = 0$$

substitute $i_o = \frac{V_1 - V_2}{10}$

$$\frac{V_2 - V_1}{10} - \frac{4(V_1 - V_2)}{10} + \frac{V_2}{20} = 0$$

$$2V_2 - 2V_1 - 8V_1 + 8V_2 + V_2 = 0$$

$$-10V_1 + 11V_2 = 0 \qquad (2)$$

from 1 and 2

$V_1 = 176$ V $\quad V_2 = 160$ V

$$V_{o2} = V_1 - V_2 \qquad i_{o2} = \frac{V_1 - V_2}{10}$$

$$= 16 \text{ V} \qquad = 1.6 \text{ A}$$

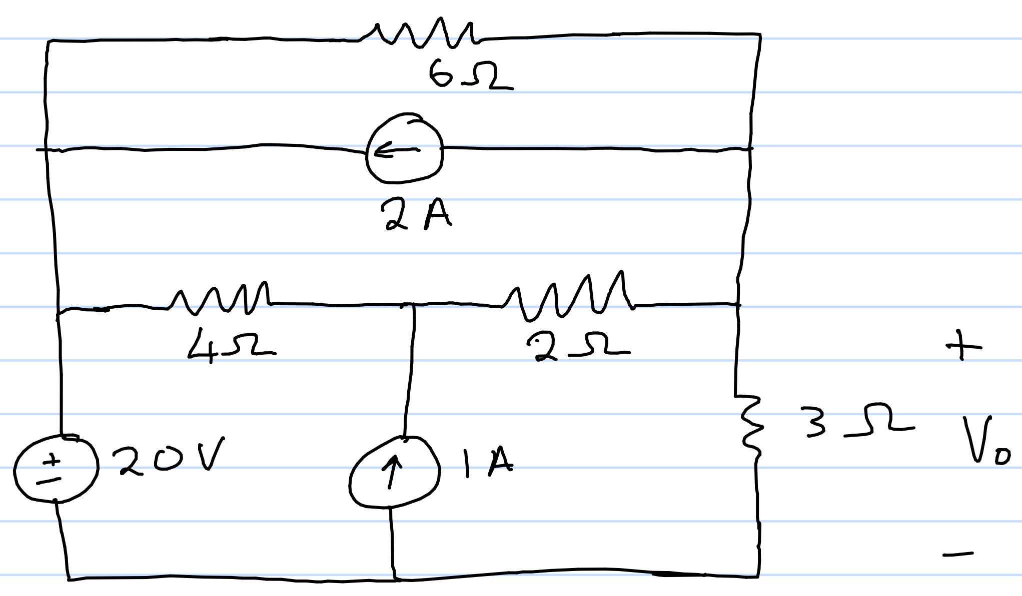
Apply superposition

$$V_o = V_{o1} + V_{o2} \qquad i_o = i_{o1} + i_{o2}$$

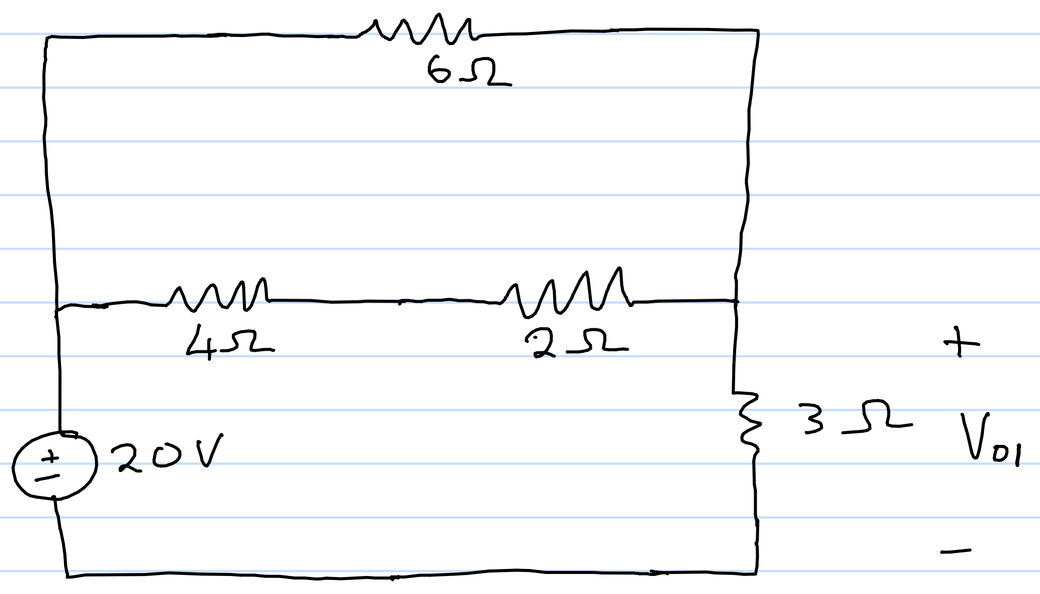
$$= 2 + 16 \qquad = 0.2 + 1.6$$

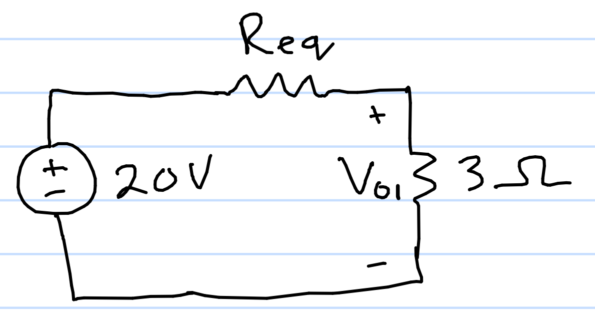
$$= 18 \text{ V} \qquad = 1.8 \text{ A}$$

4.14 Superposition (find V_o)



Keep 20 V source





$R_{eq} = (4+2) \| 6$

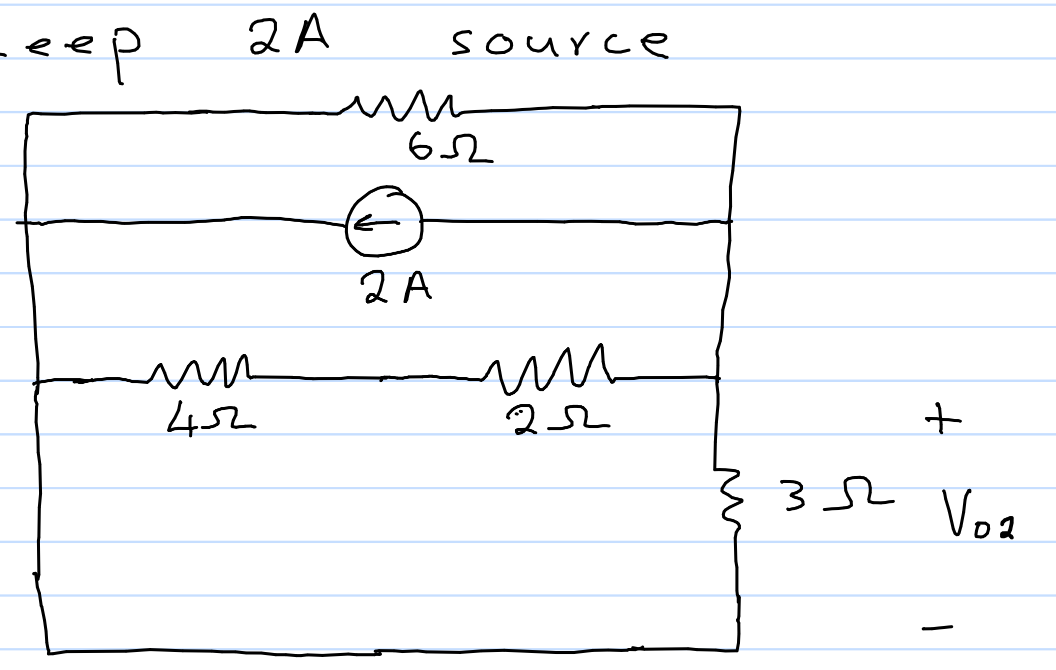
$= 3\ \Omega$

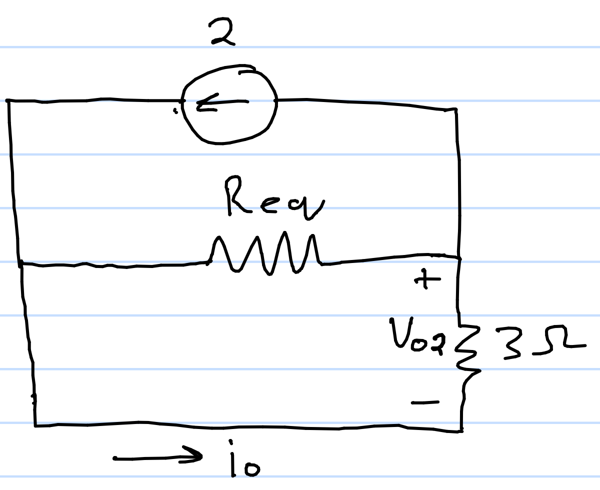
Voltage division

$V_{o1} = \dfrac{20 \times 3}{3+3}$

$= 10\ V$

Keep 2A source





$R_{eq} = (4+2) \| 6$

$= 3\ \Omega$

Current division for i_o

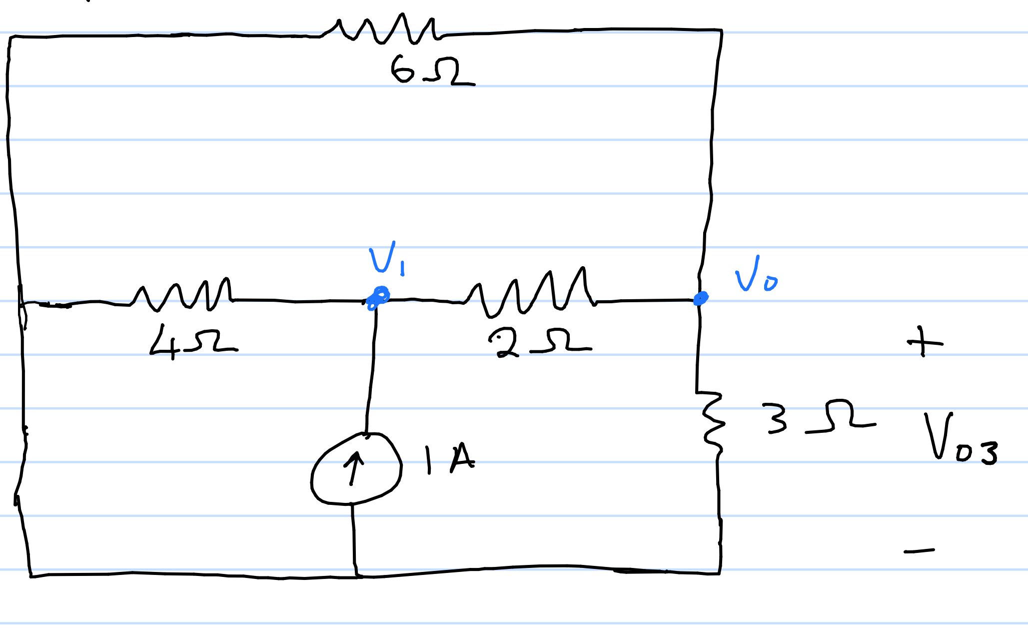
$i_o = \dfrac{2 \times 3}{3+3}$

$= 1\ A$

$V_{o2} = -1 \times 3$

$= -3\ V$

Keep 1 A source



Perform nodal analysis by inspection

$G_{00} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$ $\qquad G_{01} = -\frac{1}{2}$

$G_{10} = -\frac{1}{2}$ $\qquad G_{11} = \frac{1}{2} + \frac{1}{4}$

$$\begin{bmatrix} \frac{1}{1} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} V_{o3} \\ V_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$V_{o3} = 1, \quad V_1 = 2$

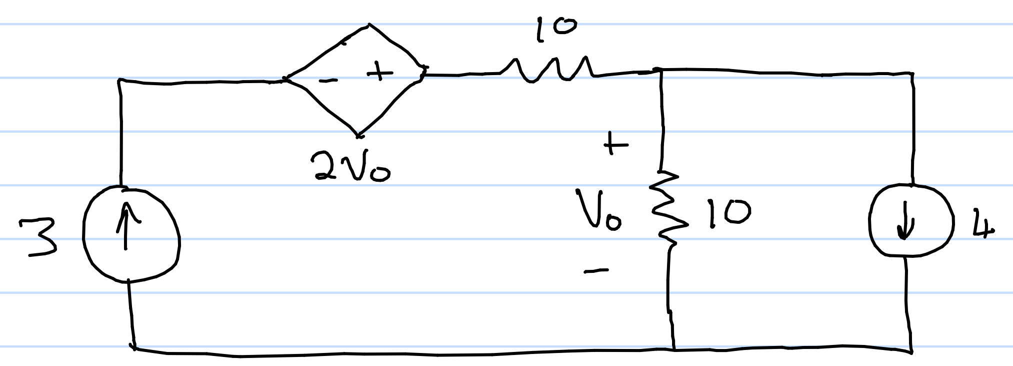
Apply superposition

$V_o = V_{o1} + V_{o2} + V_{o3}$

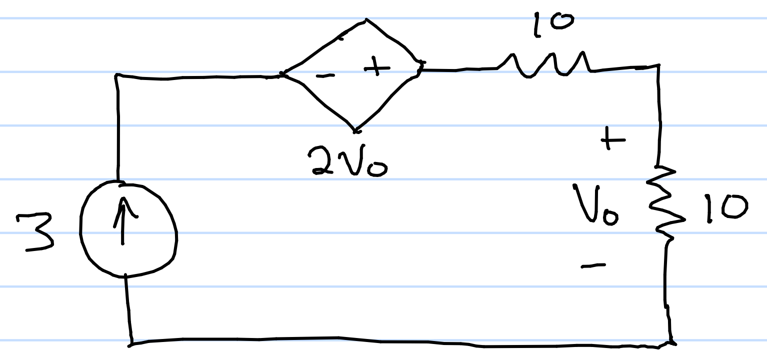
$= 10 - 3 + 1$

$= 8\ V$

4.18 Superposition (find V_o)



Keep 3A source

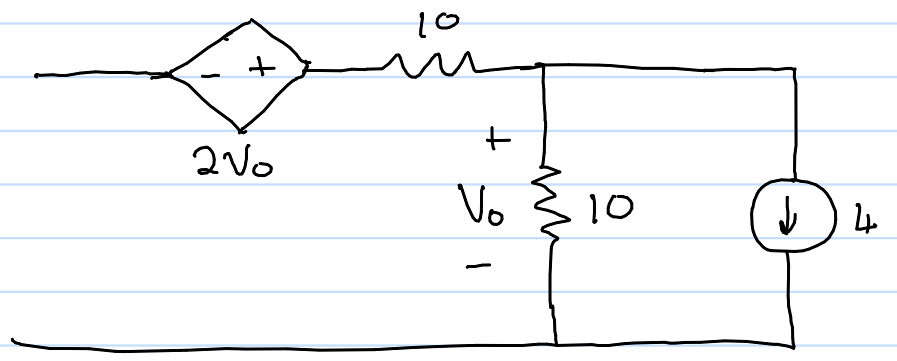


Everything is in series

Apply ohm's law

$V_{o1} = 3 \times 10$
$= 30 \text{ V}$

Keep 4A source



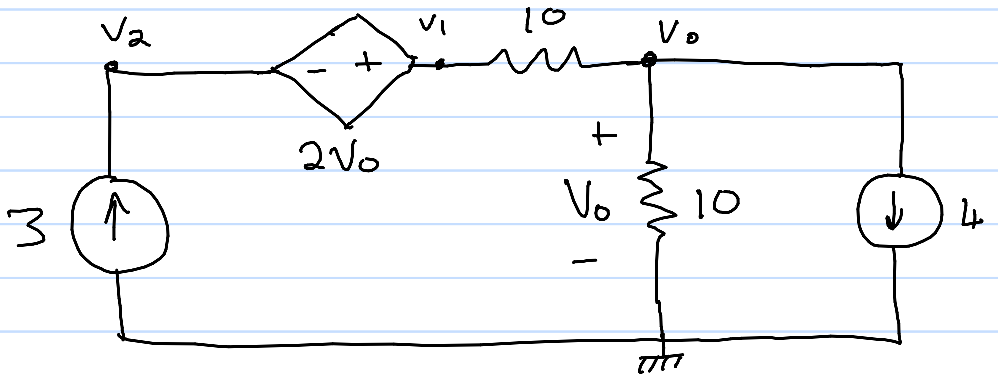
Ohm's law

$V_{o2} = -4 \times 10$
$= -40 \text{ V}$

Apply superposition

$V_o = V_{o1} + V_{o2}$
$= 30 - 40$
$= -10 \text{ V}$

Check answer using nodal analysis



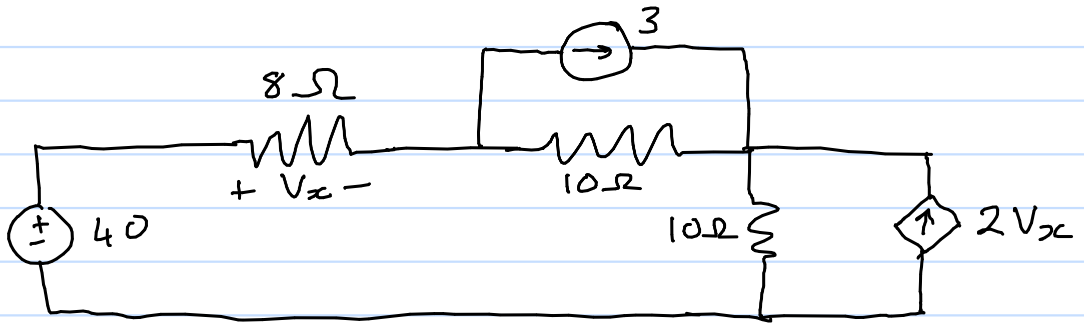
@ Node V_o

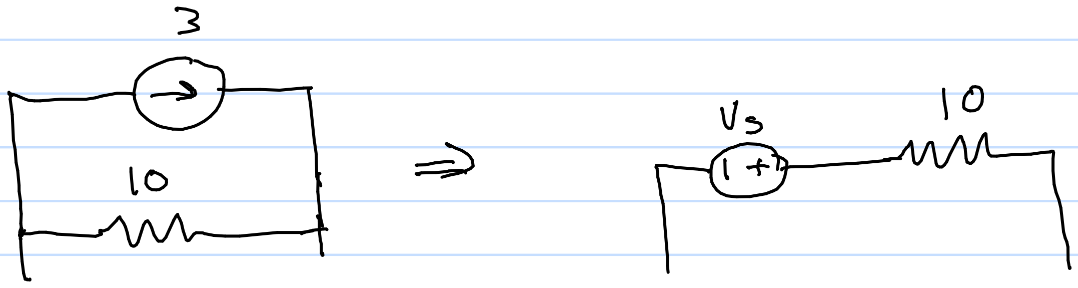
$$3 = \frac{V_o}{10} + 4$$

$$30 = V_o + 40$$

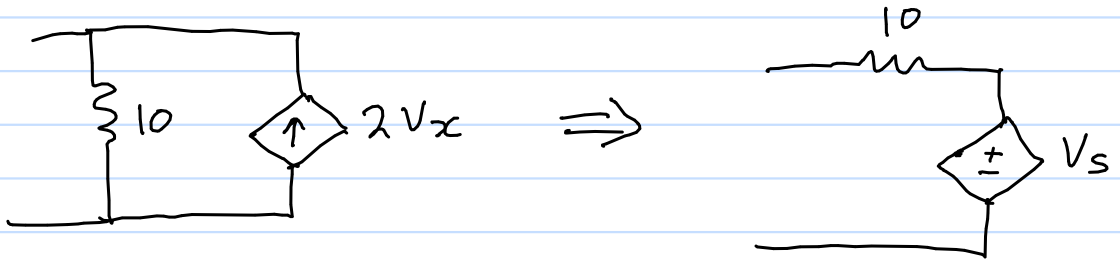
$$V_o = -10 \text{ V}$$

4.24 Source Transformation (find V_x)

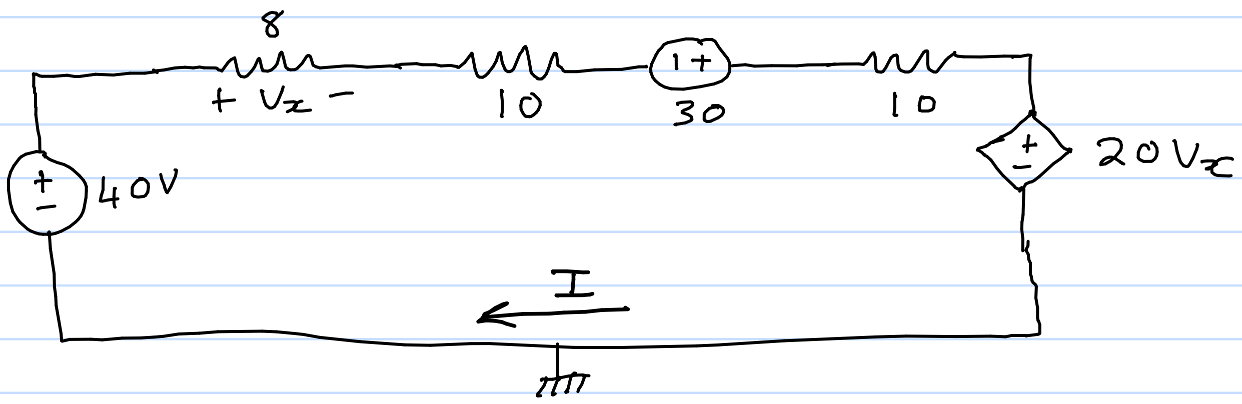




$$V_s = I_s R$$
$$= 3 \times 10$$
$$= 30\,V$$



$$V_s = I_s R$$
$$= 2V_x \times 10$$
$$= 20V_x$$



Apply KVL

$$-40 + V_x + 10I - 30 + 10I + 20V_x = 0$$

$$21V_x + 20I = 70 \qquad (1)$$

Apply Ohm's law at V_x

$$V_x = 8I$$

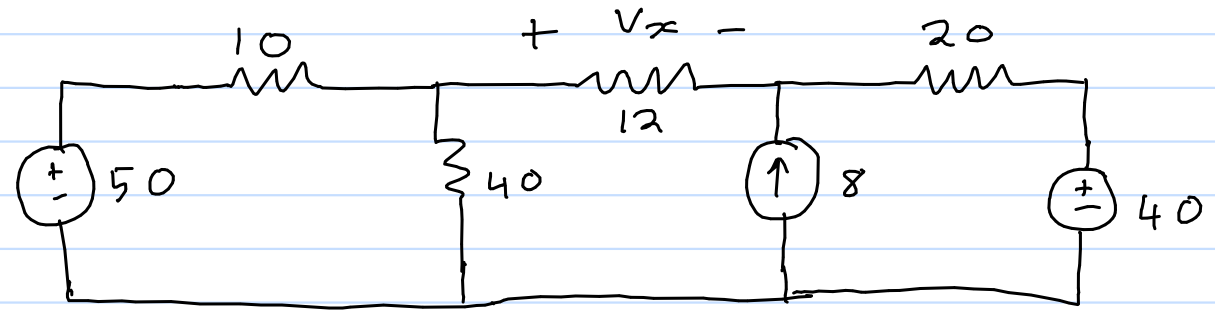
$$V_x - 8I = 0 \qquad (2)$$

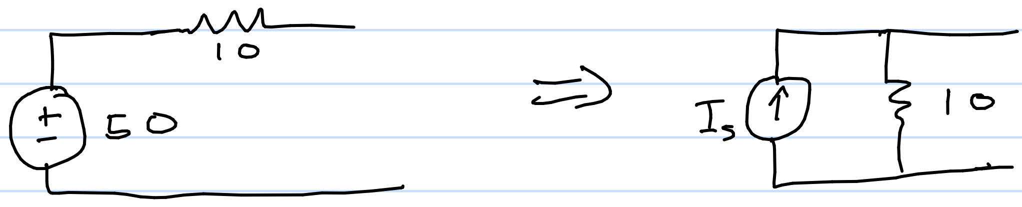
$$\begin{bmatrix} 21 & 20 \\ 1 & -8 \end{bmatrix} \begin{bmatrix} V_x \\ I \end{bmatrix} = \begin{bmatrix} 70 \\ 0 \end{bmatrix}$$

$$V_x = 2.979\ V$$

$$I = 0.372\ A$$

4.27 Source Transformation (find V_x)

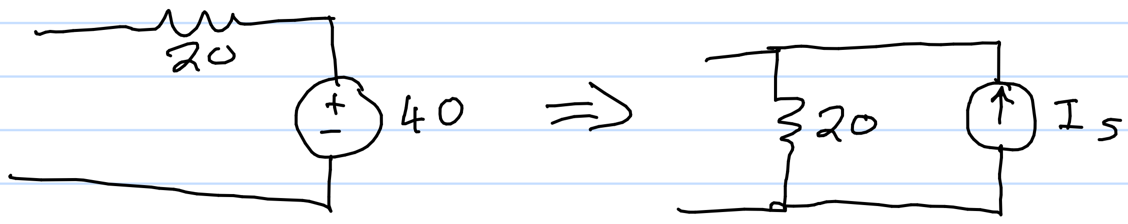




$$I_s = \frac{V_s}{R}$$

$$= \frac{50}{10}$$

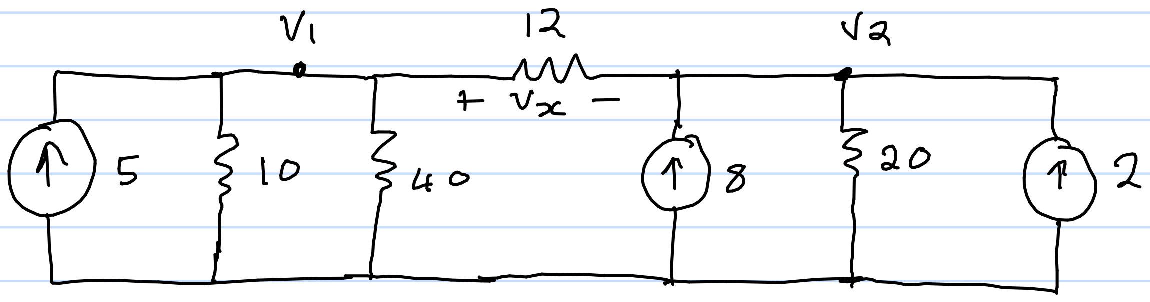
$$= 5\text{ A}$$



$$I_s = \frac{V_s}{R}$$

$$= \frac{40}{20}$$

$$= 2\text{ A}$$



$V_x = V_1 - V_2$

Nodal analysis by inspection

$G_{11} = \frac{1}{10} + \frac{1}{40} + \frac{1}{12}$ $\quad G_{12} = \frac{-1}{12}$

$G_{21} = \frac{-1}{12}$ $\quad G_{22} = \frac{1}{12} + \frac{1}{20}$

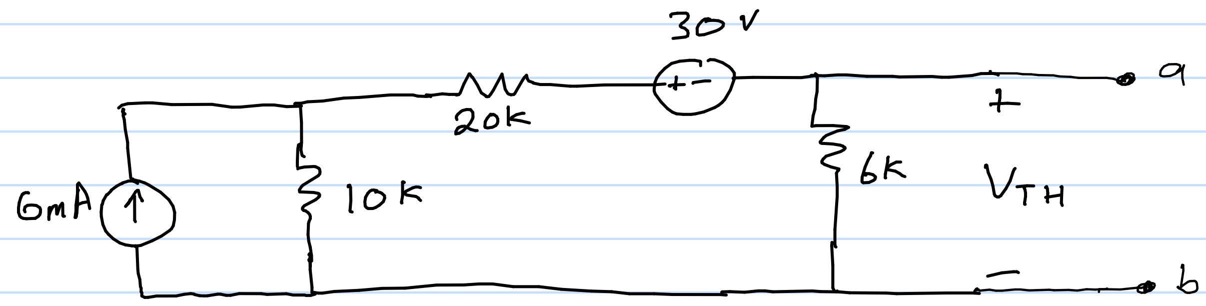
$$\begin{bmatrix} \frac{5}{24} & \frac{-1}{12} \\ \frac{-1}{12} & \frac{2}{15} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$V_1 = 72$

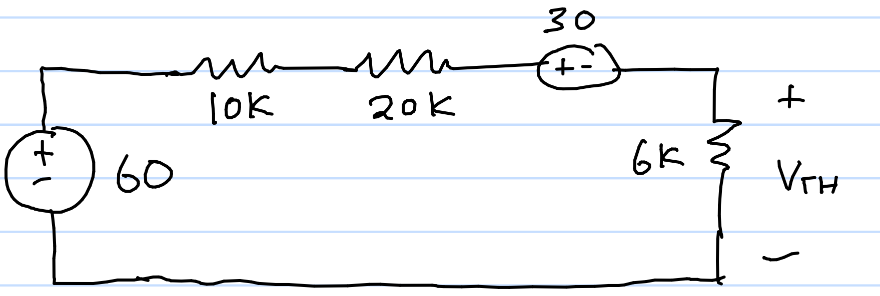
$V_2 = 120$

$V_x = V_1 - V_2 = -48\text{ V}$

4.37 Thevenin equivalent
(across a-b)



Apply source transformation



$V_s = I_s R$
$= 6mA \times 10K\Omega$
$= 60V$

Apply KVL

$$-60 + 30000I + 30 + V_{TH} = 0$$

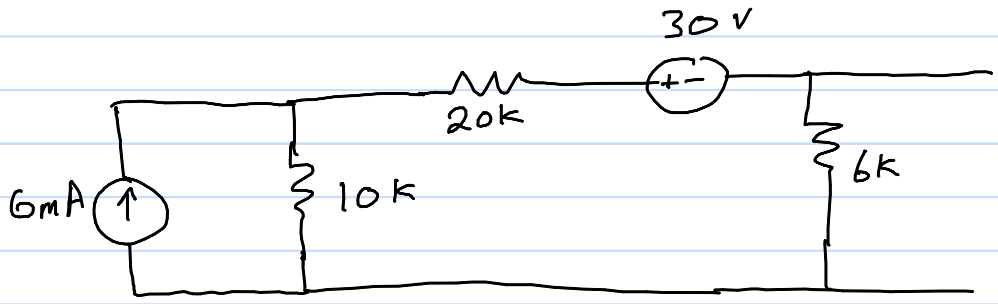
$$30000I + V_{TH} = 30 \quad (1)$$

$$V_{TH} = 6000I$$

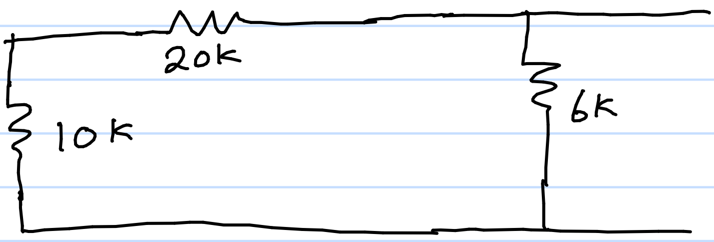
$$-6000I + V_{TH} = 0 \quad (2)$$

$$\begin{bmatrix} 30000 & 1 \\ -6000 & 1 \end{bmatrix} \begin{bmatrix} I \\ V_{TH} \end{bmatrix} = \begin{bmatrix} 30 \\ 0 \end{bmatrix}$$

$I = 833{,}33\ \mu A$
$V_{TH} = 5\ V$



Set all independent sources to 0



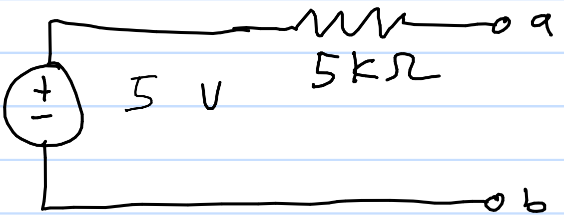
$\Rightarrow R_{TH}$

$$R_{TH} = (10K + 20K) \| 6K$$

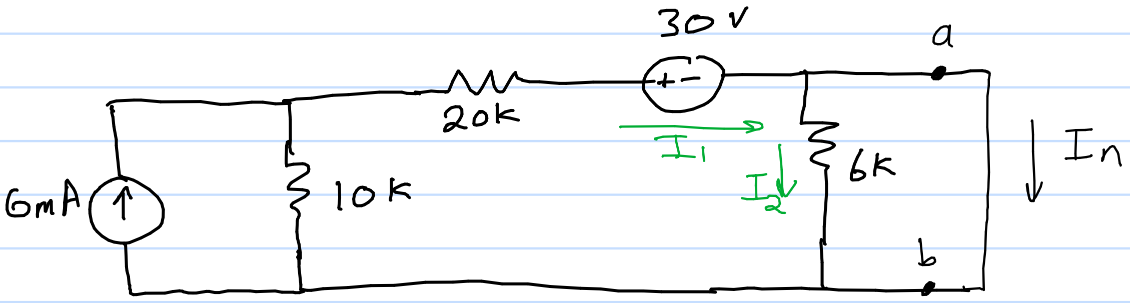
$$= \frac{6000 \times 30000}{6000 + 30000}$$

$$= 5000\ \Omega$$

Equivalent cicuit



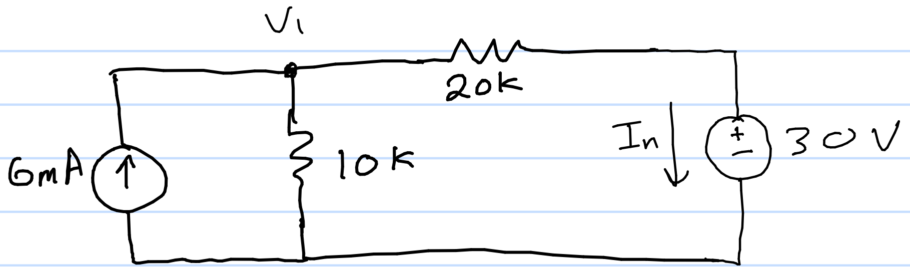
4.37 Norton Equivalent (New)
(across a-b)



Through current division

$I_2 = \frac{I_1 \times 0}{6K + 0} = 0$ $\qquad$ $I_n = \frac{I_1 \times 6K}{6K + 0} = I_1$

Redraw circuit



Apply KCL at V_1

$$6 \times 10^{-3} = \frac{V_1}{10 \times 10^3} + \frac{V_1 - 30}{20 \times 10^3}$$

$$6 = \frac{V_1}{10} + \frac{V_1 - 30}{20}$$

$$120 = 2V_1 + V_1 - 30$$

$$150 = 3V_1$$

$$V_1 = 50 \text{ V}$$

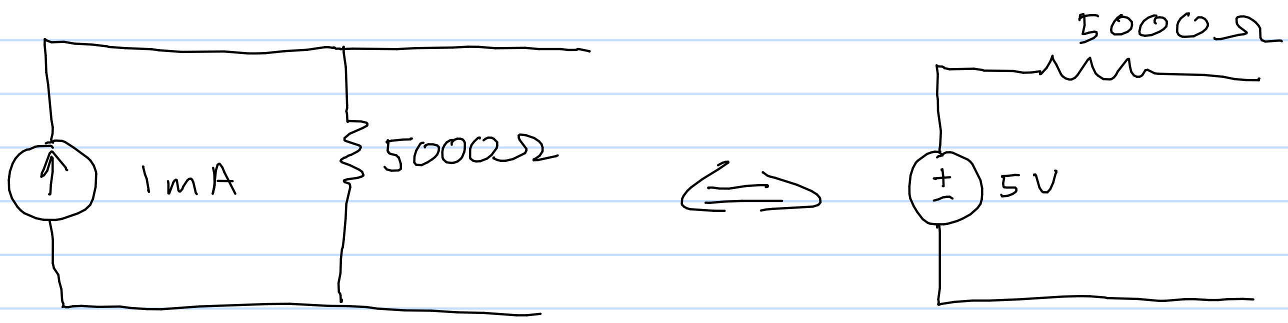
$$I_n = \frac{V_1 - 30}{20 \times 10^3}$$

$$I_n = \frac{50 - 30}{20 \times 10^3}$$

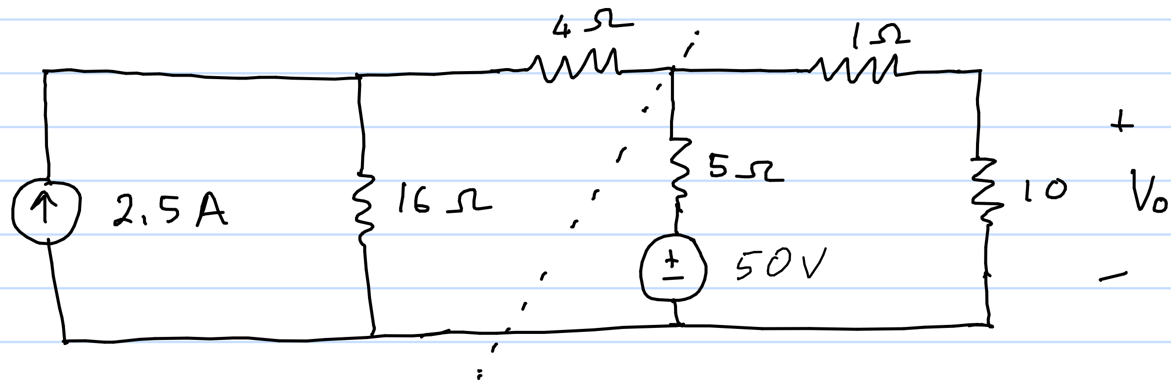
$$I_n = 1 \text{ mA}$$

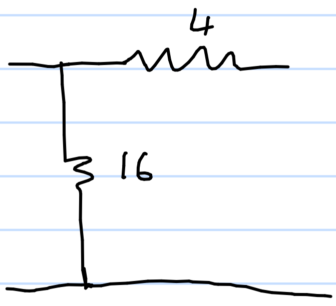
$$R_N = R_{TH} = 5000\ \Omega$$

Equivalent circuit

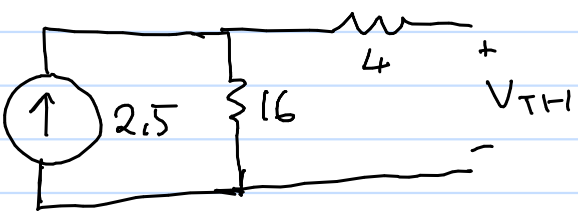


4.38 Thevenin's theorem, find V_o

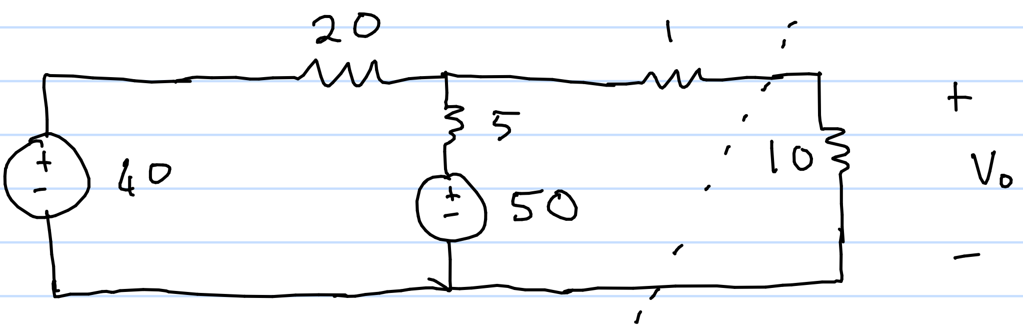




$R_{TH} = 20\,\Omega$



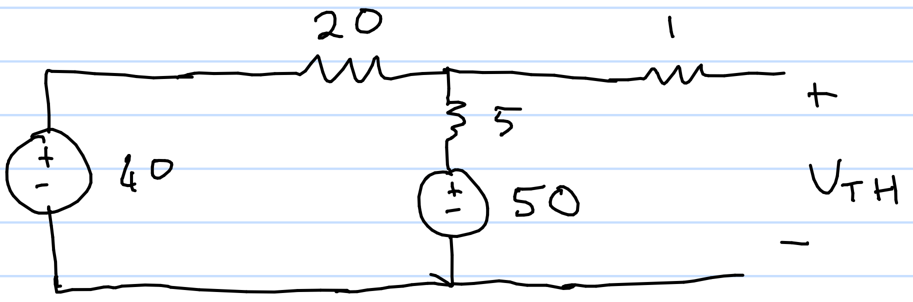
$V_{TH} = 2.5 \times 16$
$= 40\,V$



$R_{TH} = 20 \| 5 + 1$

$= \frac{20 \times 5}{20+5} + 1$

$= 5\,\Omega$

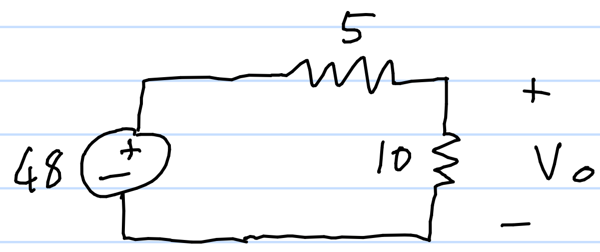


Apply KVL

$-40 + 20I + 5I + 50 = 0$
$25I = -10$
$I = -0.4\,A$

$-V_{TH} + 5I + 50 = 0$
$V_{TH} = 50 + 5I$
$= 50 - 5 \times 0.4$
$= 48\,V$

Find V_o



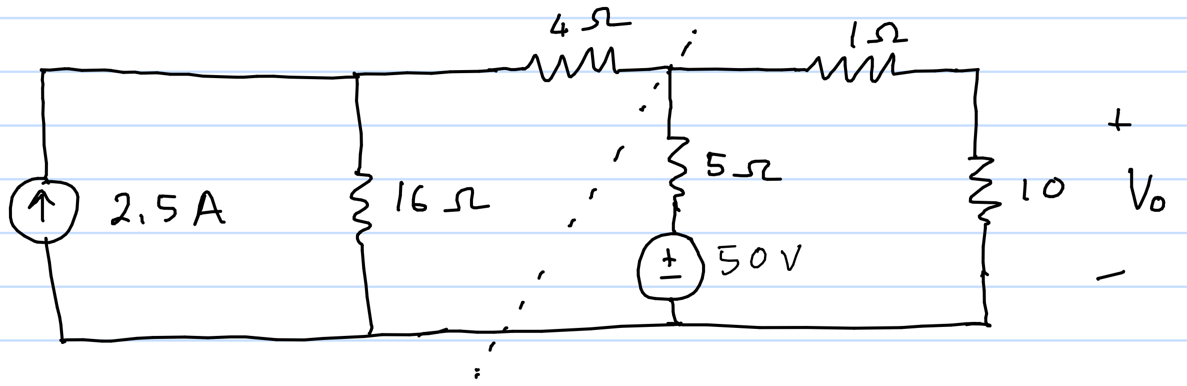
Apply voltage division

$V_o = \frac{48 \times 10}{5 + 10}$

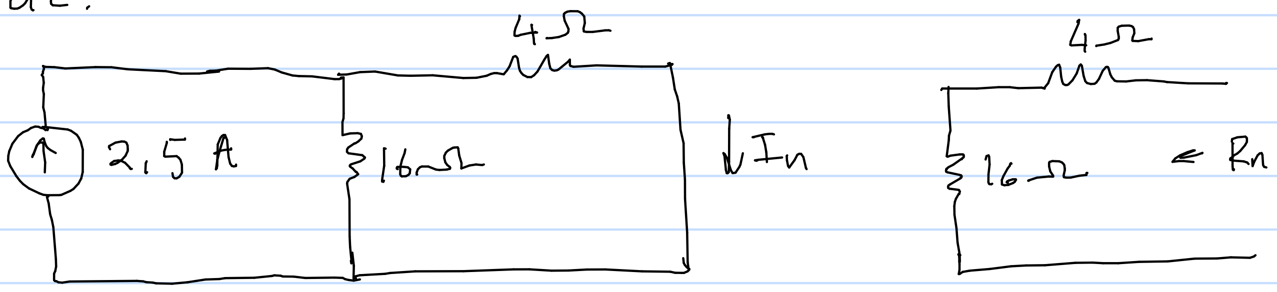
$= 32\,V$

Note: We have avoided nodal/mesh analysis by using Thevenin's theorem twice

4.38 Norton's theorem, find V_o (New)

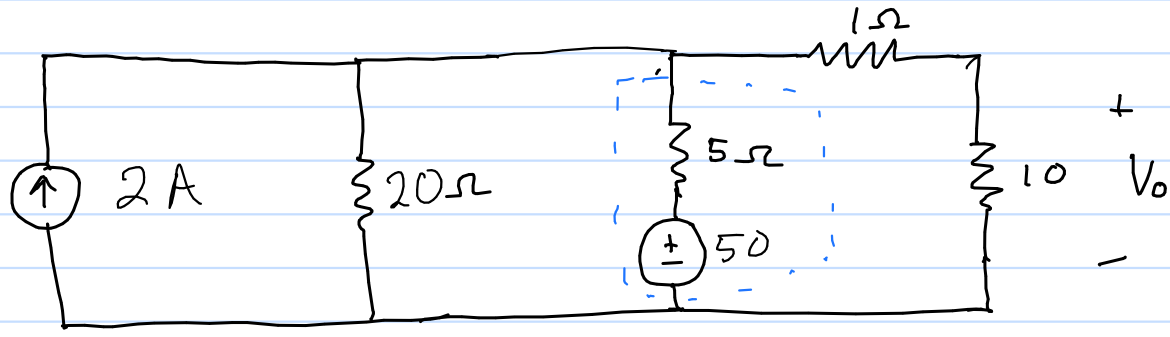


Find Norton equivalent to simplify left side.

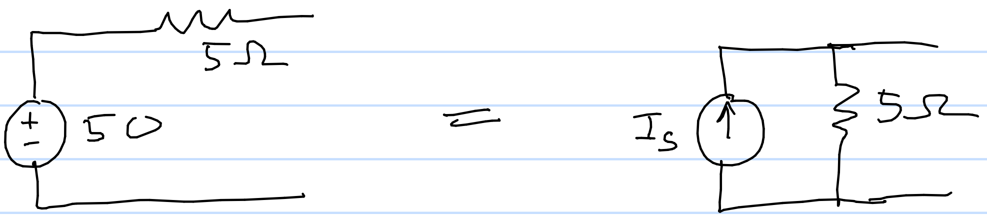


$$I_n = \frac{2.5 \times 16}{16 + 4} = 2\,A$$ (Current division)

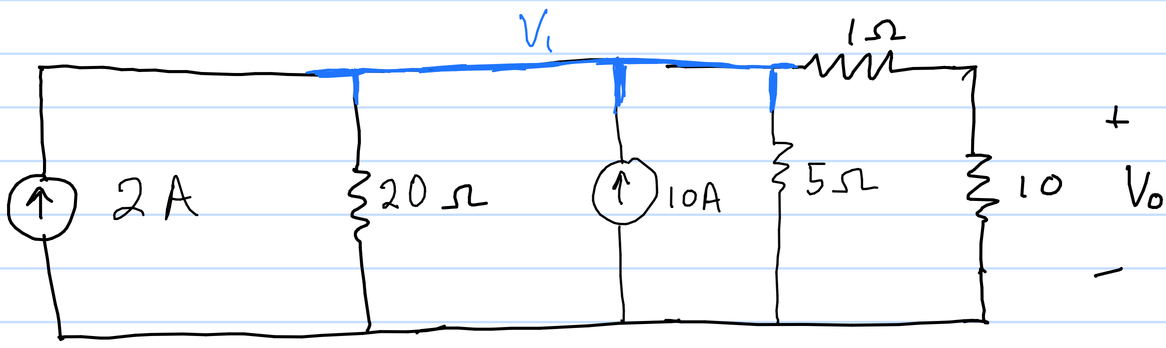
$$R_n = 16 + 4 = 20\,\Omega$$ (Series)



Source transformation



$$I_s = \frac{50}{5} = 10\,A$$



Using nodal analysis

$$\frac{V_1}{20} + \frac{V_1}{5} + \frac{V_1}{11} = 2 + 10 \qquad (\times 80)$$

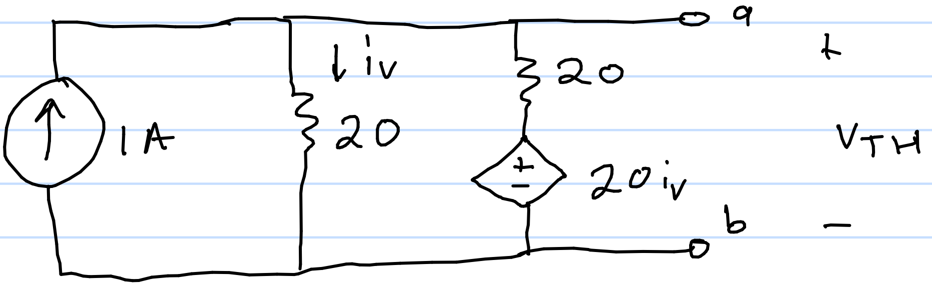
$$V_1\left(\frac{1}{20} + \frac{1}{5} + \frac{1}{11}\right) = 12$$

$$V_1 = \left(\frac{1}{20} + \frac{1}{5} + \frac{1}{11}\right)^{-1} \times 12$$

$$V_1 = 35.2\,V$$

$$V_o = \frac{V_1 \times 10}{10 + 1} = 32\,V$$ (Voltage division)

4.47 Thevenin equivalent (across a-b)



KCL at node V_{TH}

$$-1 + \frac{V_{TH}}{20} + \frac{V_{TH} - 20 i_v}{20} = 0$$

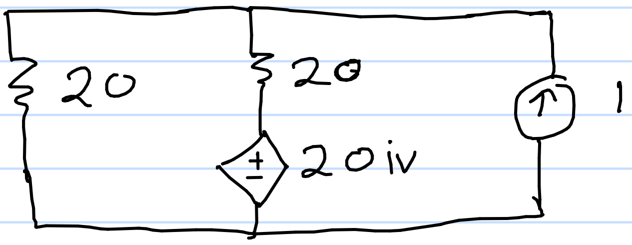
$$V_{TH} + V_{TH} - 20 i_v = 20$$

$$i_v = \frac{V_{TH}}{20}$$

$$V_{TH} + V_{TH} - 20 \frac{V_{TH}}{20} = 20$$

$$V_{TH} = 20\,V$$

To find R_{TH}, turn off 1 A source and let $i_o = 1$ A



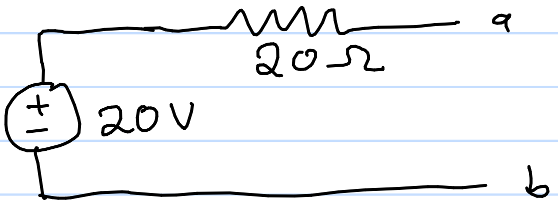
same circuit as above

$\therefore V_o = 20$ V

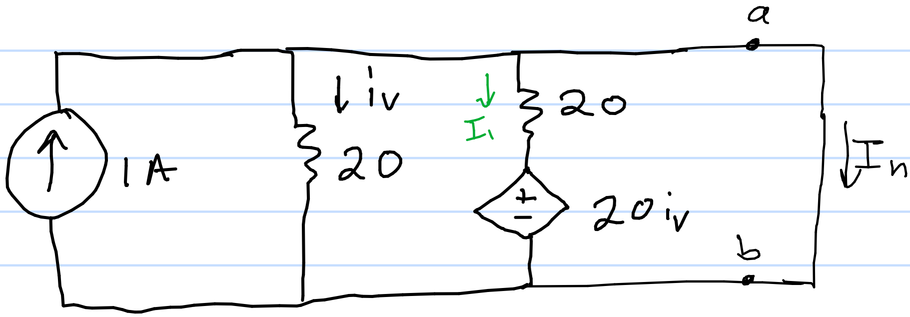
$$R_{TH} = \frac{V_o}{i_o}$$

$$= 20\,\Omega$$

equivalent circuit



4.47 Norton equivalent (New)
(across a-b)



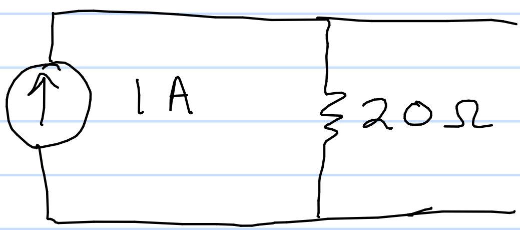
Since I_n shorts i_v and I_1

$$i_v = I_1 = 0\ A$$

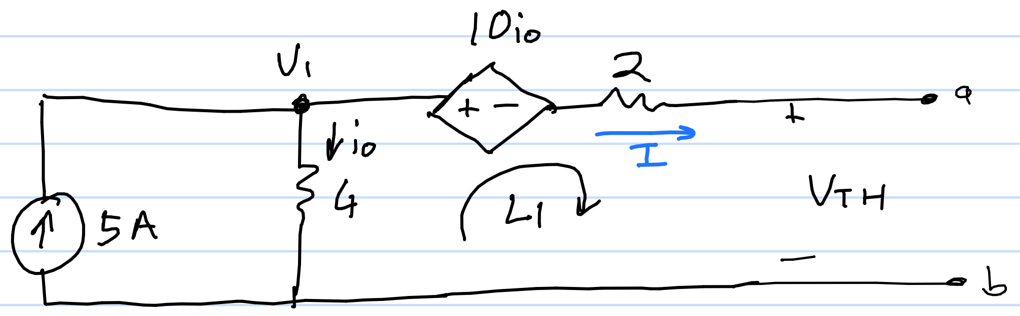
$$I_n = 1\ A$$

$$R_N = R_{TH} = 20\ \Omega$$

Equivalent circuit



4.48 Thevenin equivalent (across a-b)



KVL at Loop 1

$$-V_1 + 10i_o + 2I + V_{TH} = 0 \qquad (I = 0)$$

$$i_o = \frac{V_1}{4}$$

$$-V_1 + \frac{10V_1}{4} + V_{TH} = 0 \qquad (\times 4)$$

$$-4V_1 + 10V_1 + 4V_{TH} = 0$$

$$6V_1 + 4V_{TH} = 0$$

KCL at node V_1

$$-5 + \frac{V_1}{4} = 0$$

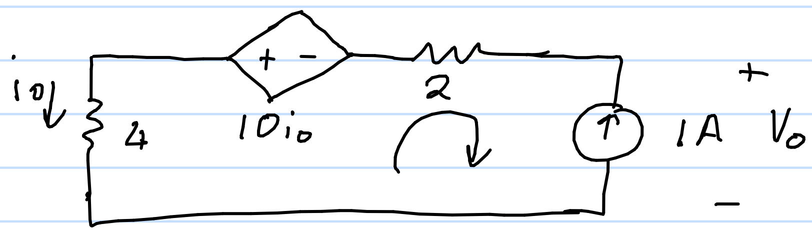
$$V_1 = 20\ V$$

$$6(20) + 4V_{TH} = 0$$

$$V_{TH} = \frac{-120}{4}$$

$$V_{TH} = -30\ V$$

Find R_{TH}, let $i_o = 1\ A$



$i_o = 1\ A$

Apply KVL

$$-4i_o + 10i_o - 2i_o + V_o = 0$$

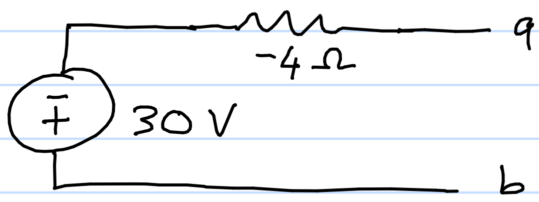
$$V_o = -4i_o$$
$$V_o = -4\ V$$

$$R_{TH} = \frac{V_o}{i_o}$$

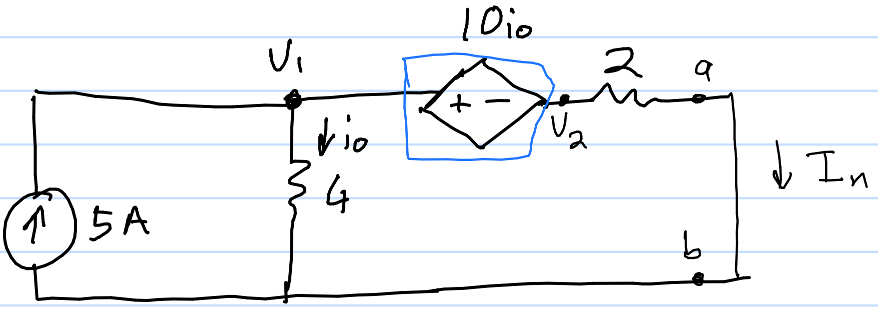
$$= \frac{-4}{1}$$

$$= -4\ \Omega$$

equivalent circuit



4.48 Norton equivalent (New)
(across a-b)



+ + $10 i_o$ − +

V_1 V_2

− −

KVL:

$$-V_1 + 10 i_o + V_2 = 0$$

$$V_2 = V_1 - 10 i_o$$

KCL at supernode

$$i_o + I_n = 5$$

$$I_n = \frac{V_1 - 10 i_o}{2}$$

$$V_1 = 4 i_o$$

$$i_o + \frac{4 i_o - 10 i_o}{2} = 5$$

$$i_o - 3 i_o = 5$$

$$i_o = \frac{5}{-2}$$

$$= -2.5 \text{ A}$$

$$I_n = \frac{4 i_o - 10 i_o}{2}$$

$$I_n = 7.5 \text{ A}$$

$$R_n = R_{TH} = -4\ \Omega$$

Equivalent circuit